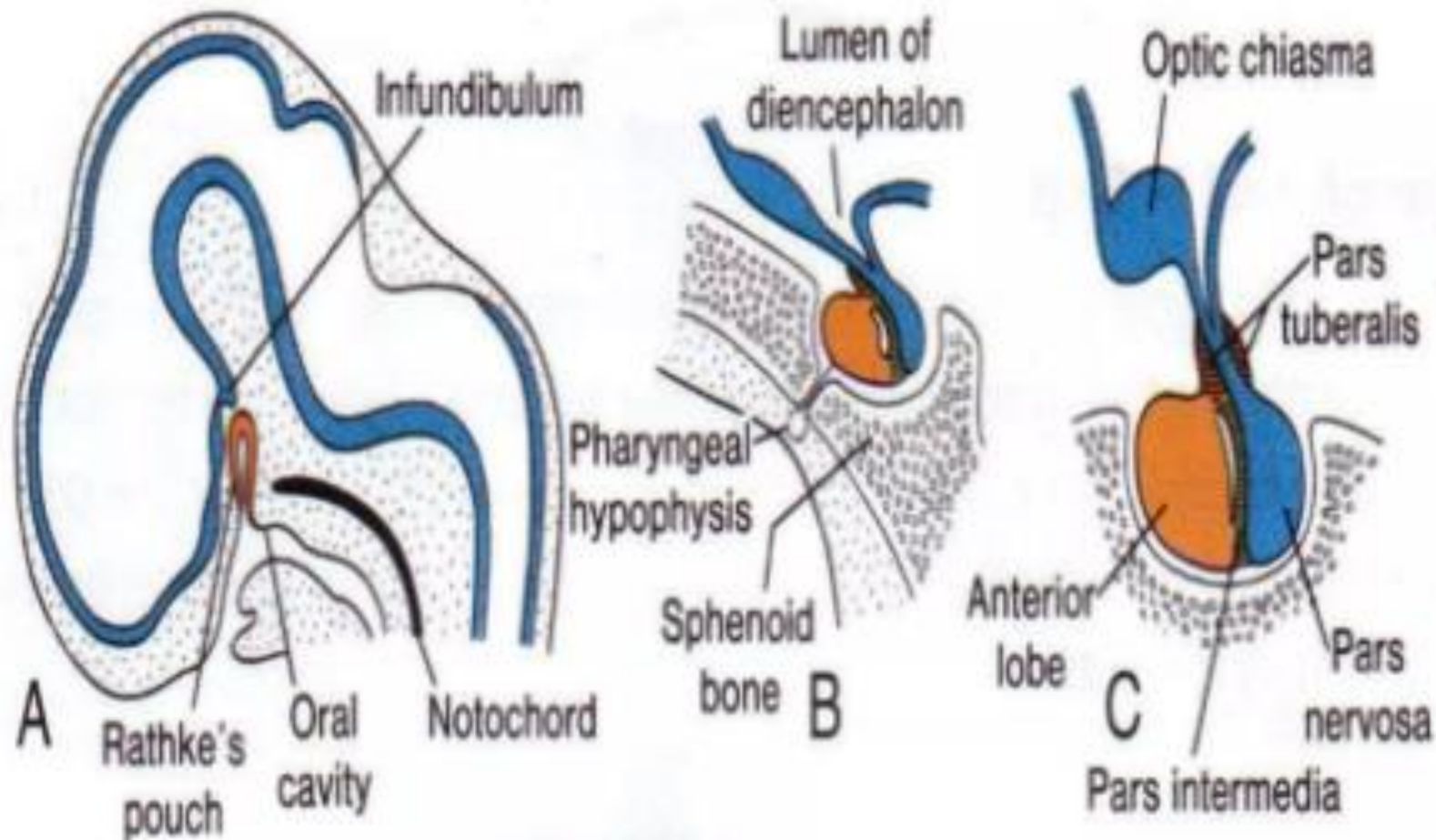


# Development of Endocrine system

**By: Fikre Bayu**

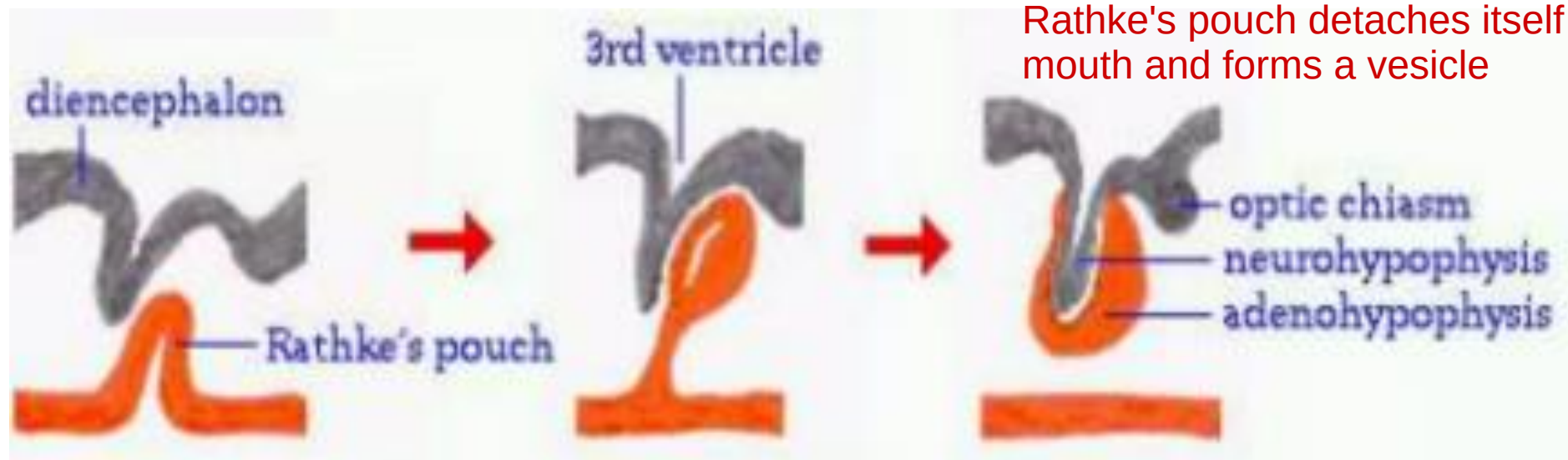
# Pituitary gland (Hypophysis cerebri)

hypophysis is a Greek word meaning attachment underneath



# Adenohypophysis Anterior pituitary

- Develops from an evagination of ectoderm lining, the roof of primitive oral cavity (stomodeum) → **Rathke's pouch....4<sup>th</sup> wk.**



Rathke's pouch detaches itself from the roof of the mouth and forms a vesicle

pars anterior =  
pars tuberalis +  
pars distalis

The cells of anterior wall of Rathke's pouch undergo extensive proliferation to form the anterior lobe (pars anterior)

The posterior wall proliferates more slowly to form the intermediate lobe (pars intermedia)

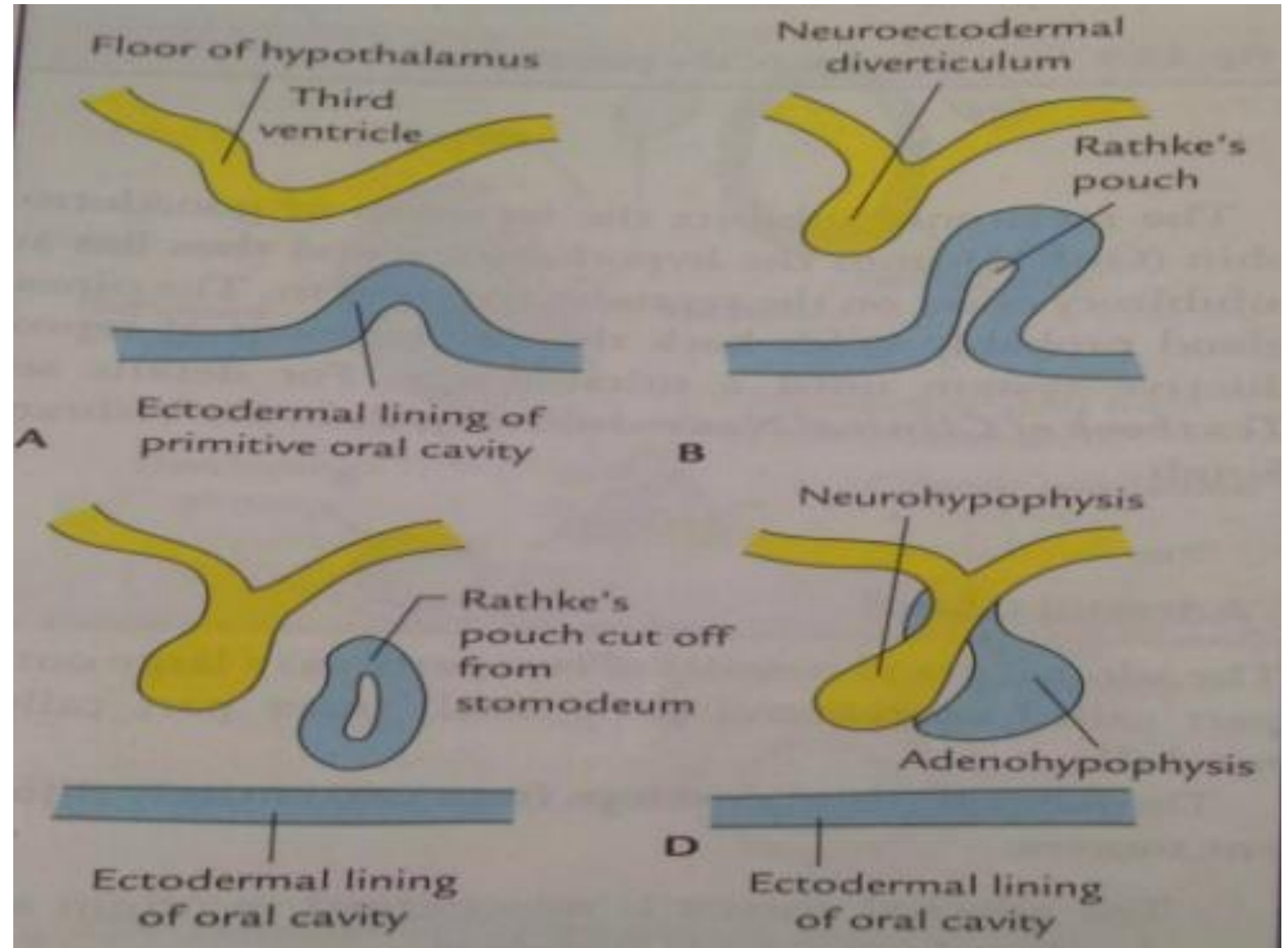
the space between pars intermedia and pars anterior is called pituitary cleft

# Neurohypophysis

posterior pituitary= pars nervosa + infundibular stalk

the infundibulum gives the posterior lobe

- Develops from an evagination of neuroectoderm of diencephalon (floor of 3<sup>rd</sup> ventricle)
- Fused with from the Rathke's pouch → Hypophysis cerebri



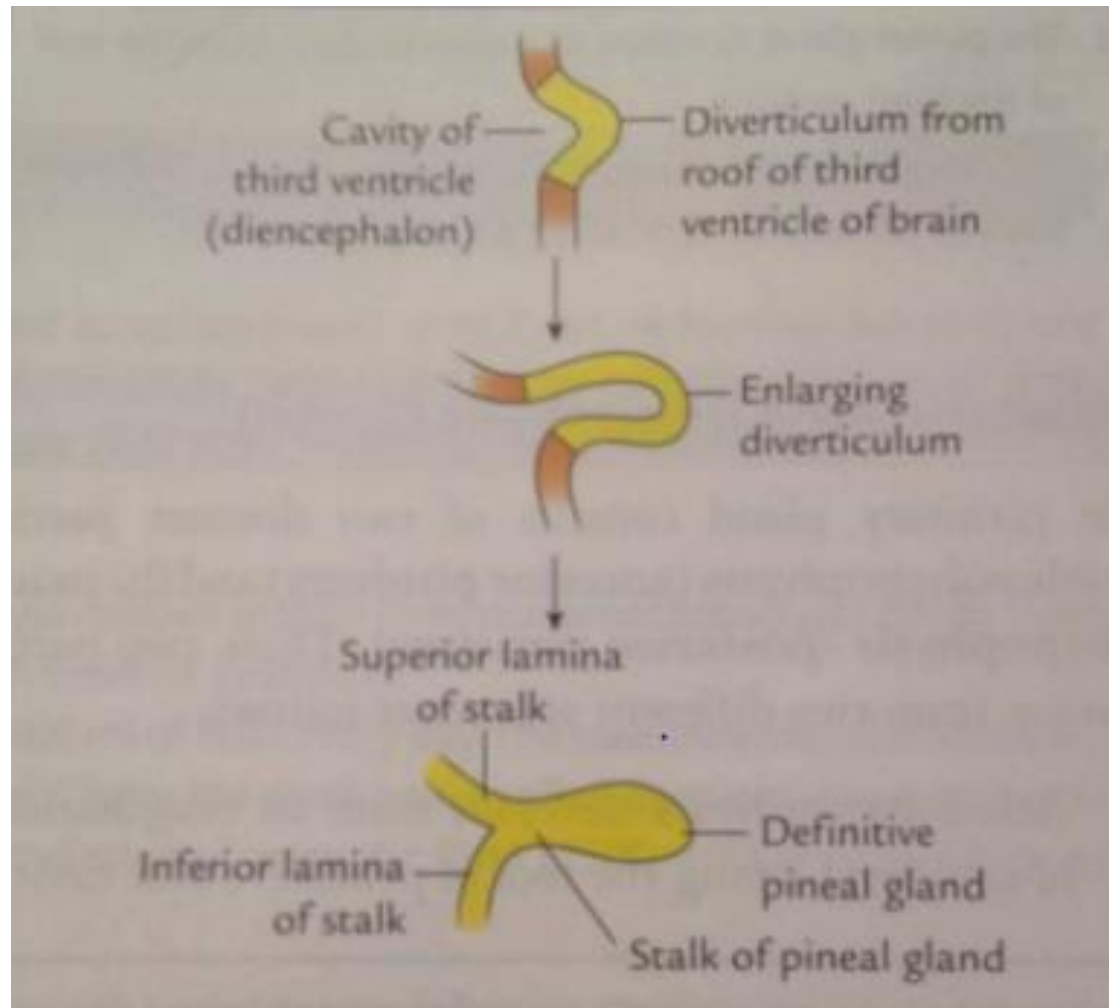
# Pineal gland (Epiphysis cerebri)

- Develops from the roof of the diencephalon

Ectopic posterior pituitary is a condition in which the back end (posterior) of the pituitary gland is in an abnormal position.

The pharyngeal hypophysis is a small piece of extra sellar pituitary tissue derived from the stalk of Rathke's pouch and is situated in the roof of the nasopharynx

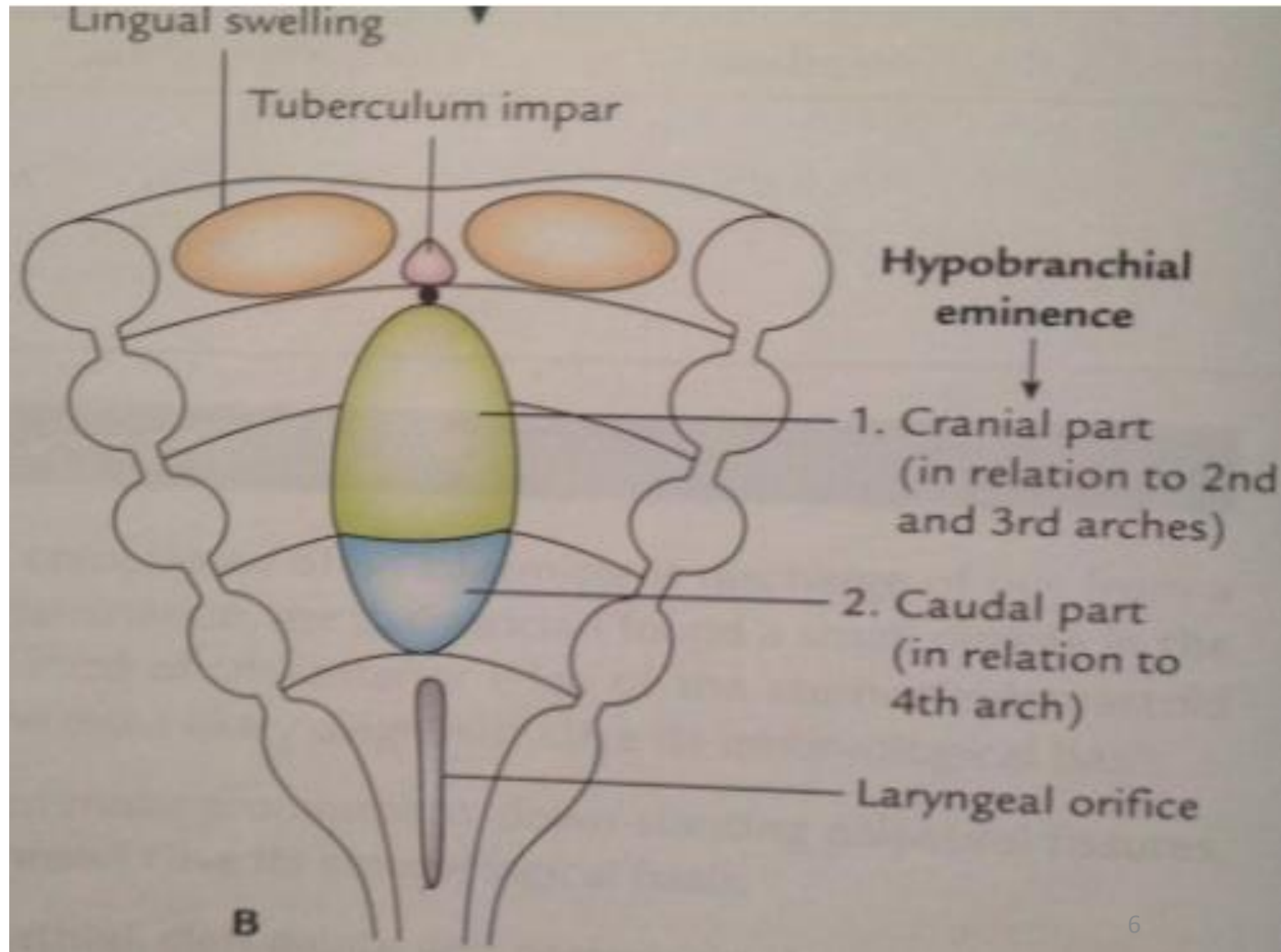
A craniopharyngioma is a rare type of brain tumor derived from pituitary gland embryonic tissue that occurs most commonly in children, but also affects adults



# Thyroid gland

- In the **midline of the floor of the pharynx**, the **endodermal lining of the foregut forms the thyroid diverticulum**
- **Migrates caudally, passing ventral to the hyoid bone and laryngeal cartilages**
- Remains **connected to the tongue by the thyroglossal duct, which later is obliterated**

8/30/2023



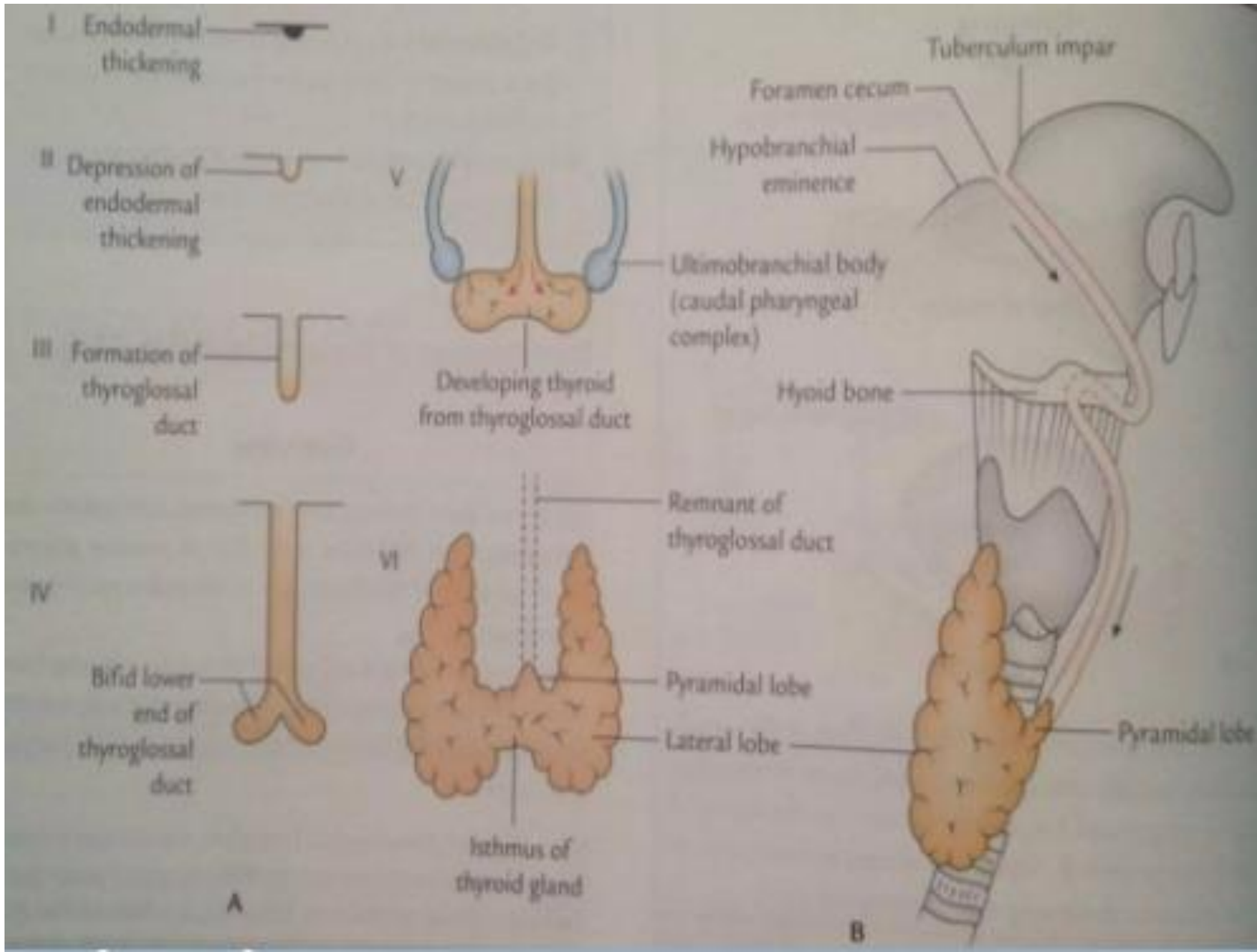
# Cont....

A thyroglossal duct cyst is a mass or lump in the front part of neck that is filled with fluid. A thyroglossal cyst is a fibrous cyst that forms from a persistent thyroglossal duct.

Thyroglossal duct originates at the junction of the anterior two-thirds and the posterior third of the tongue at its base(foramen cecum).it connects the thyroid diverticulum to the tongue

Lingual thyroid is a mass of ectopic thyroid tissue, located at the base of the tongue, caused by a developmental anomaly of the descent of thyroid gland

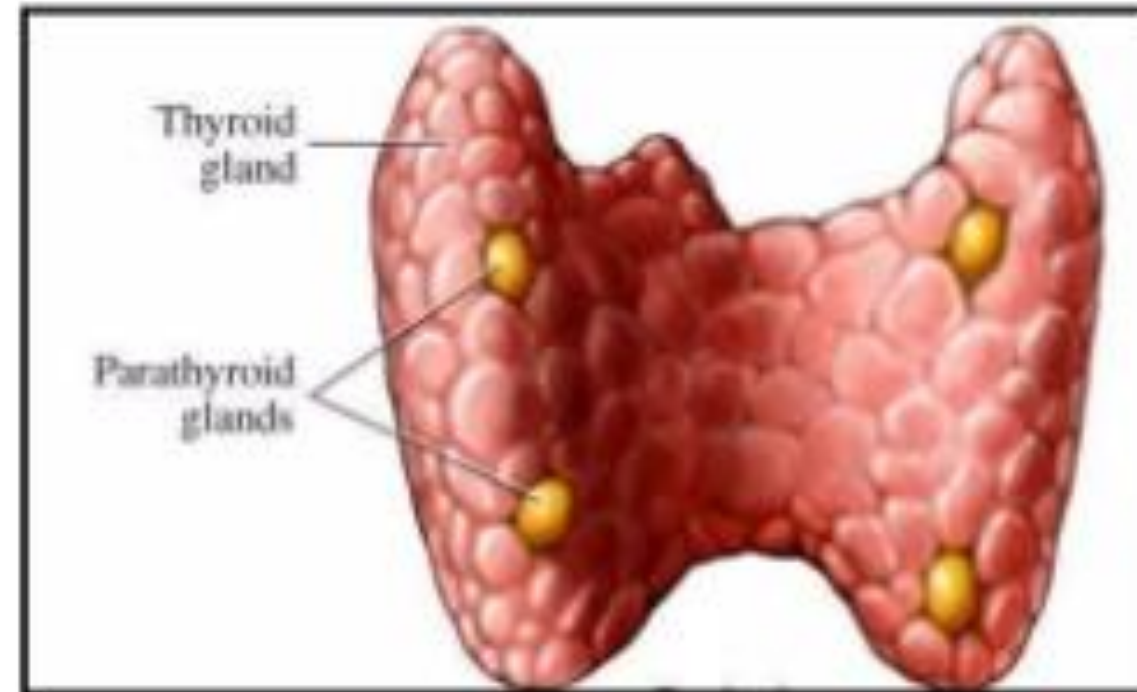
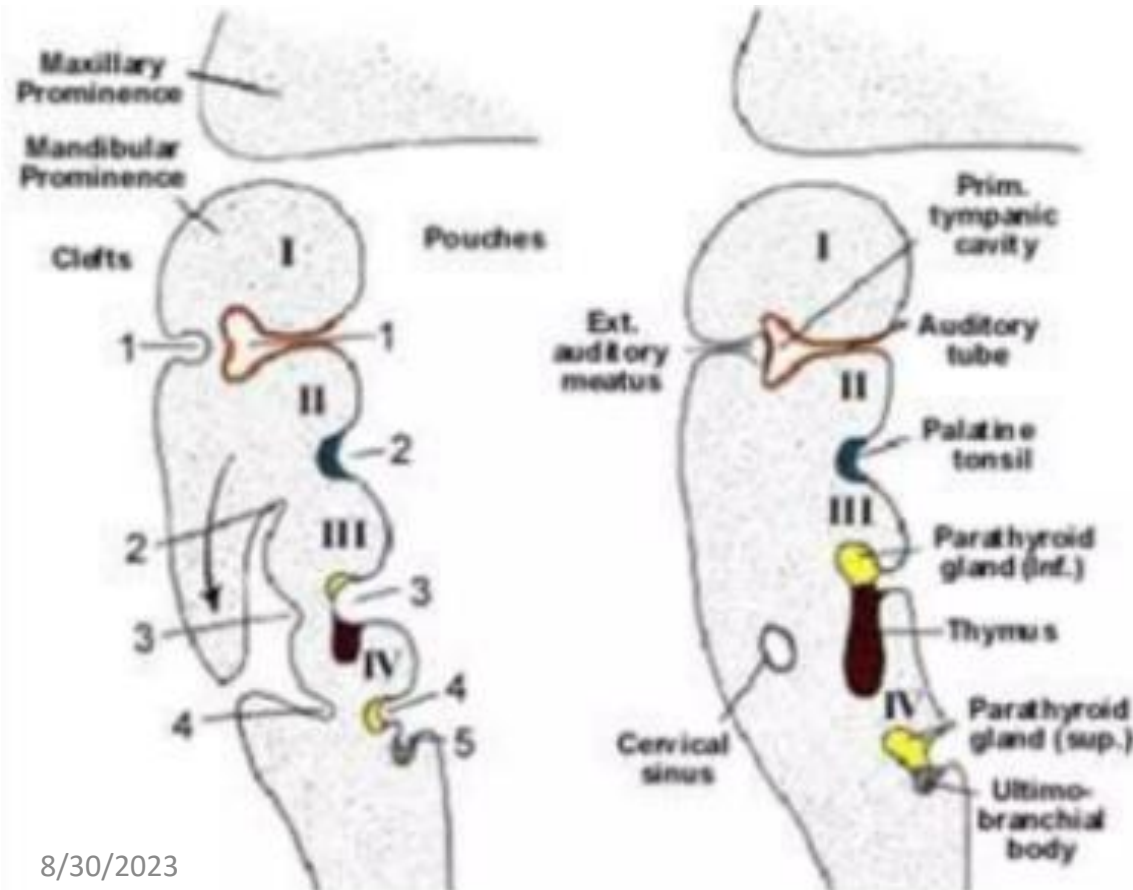
The thyroid gland is normally located at the front of the neck. A retrosternal thyroid refers to the abnormal location of all or part of the thyroid gland below the breastbone (sternum)



# Parathyroid glands

- Develop from 3<sup>rd</sup> and 4<sup>th</sup> pharyngeal pouches

The third pharyngeal pouch gives rise to the inferior parathyroid glands, while the superior parathyroids arise from the fourth pharyngeal pouch.



# Suprarenal gland

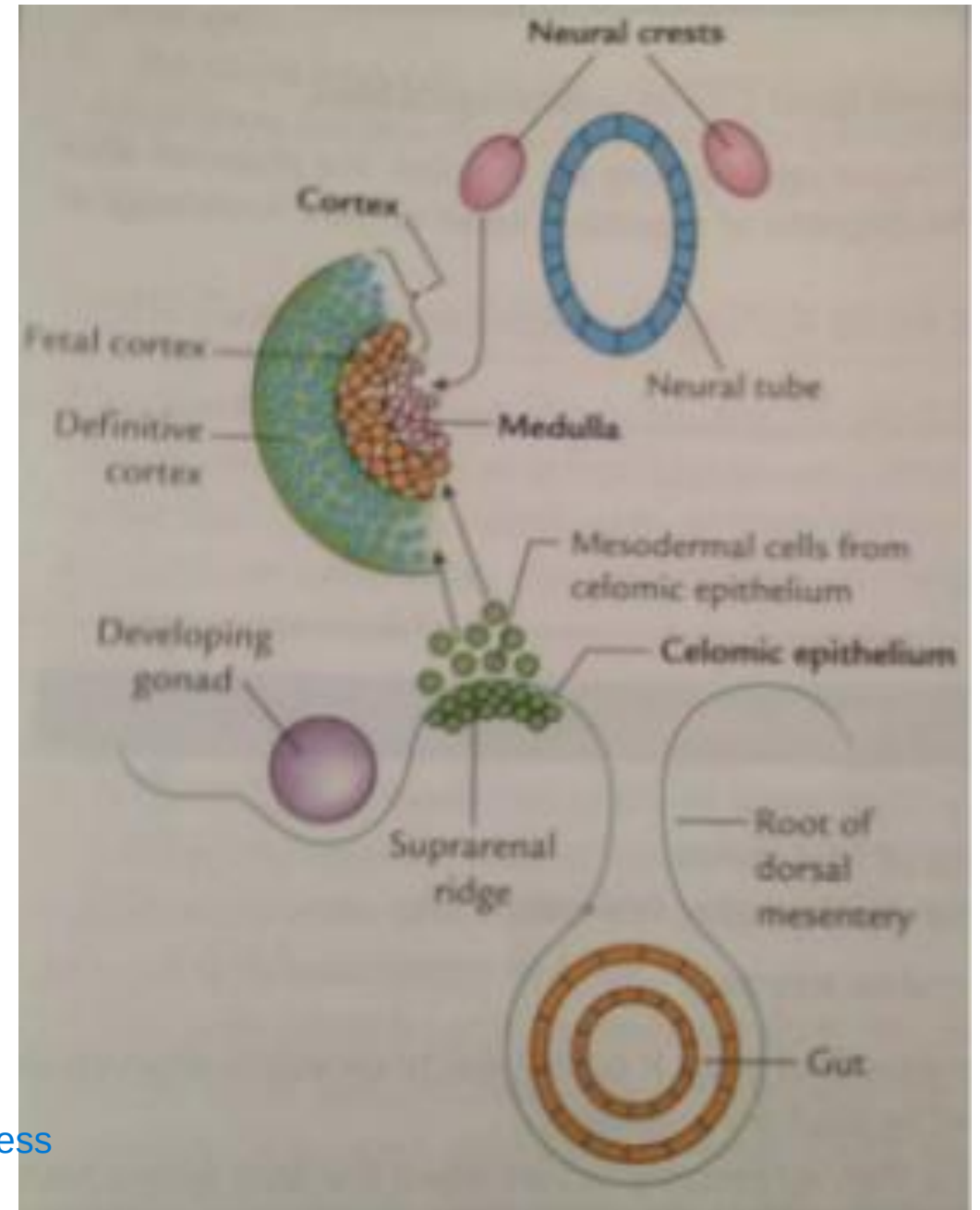
- **Cortex → coelomic epithelium**

- Fetal cortex → larger acidophilic cells
- Definitive cortex → smaller cells

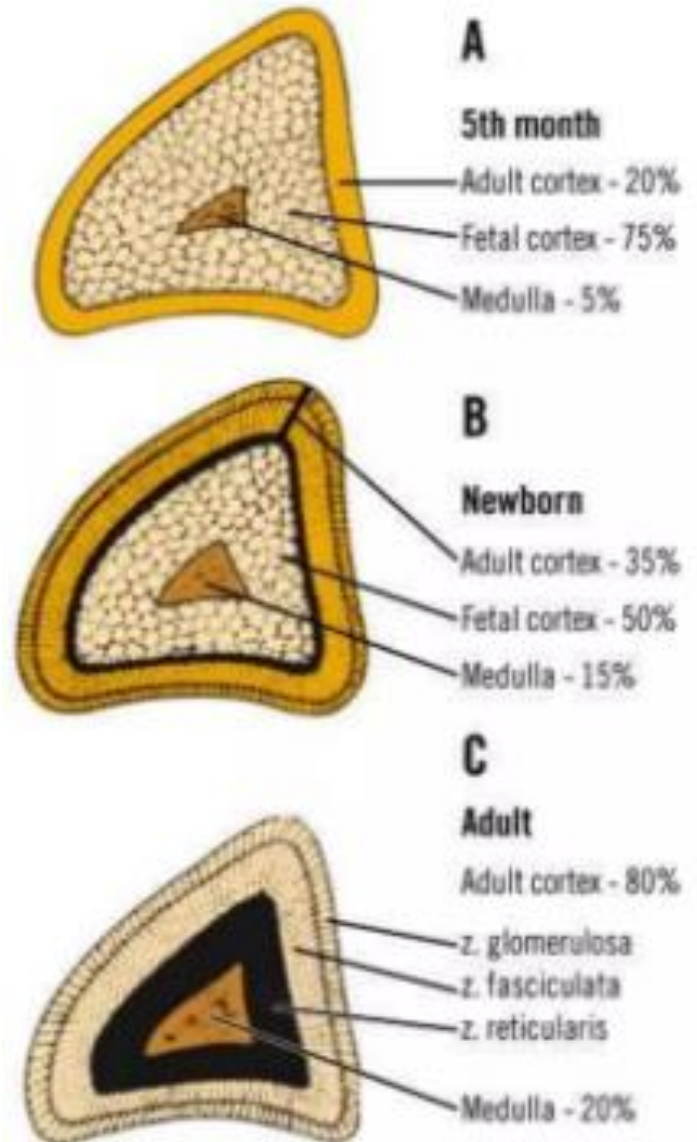
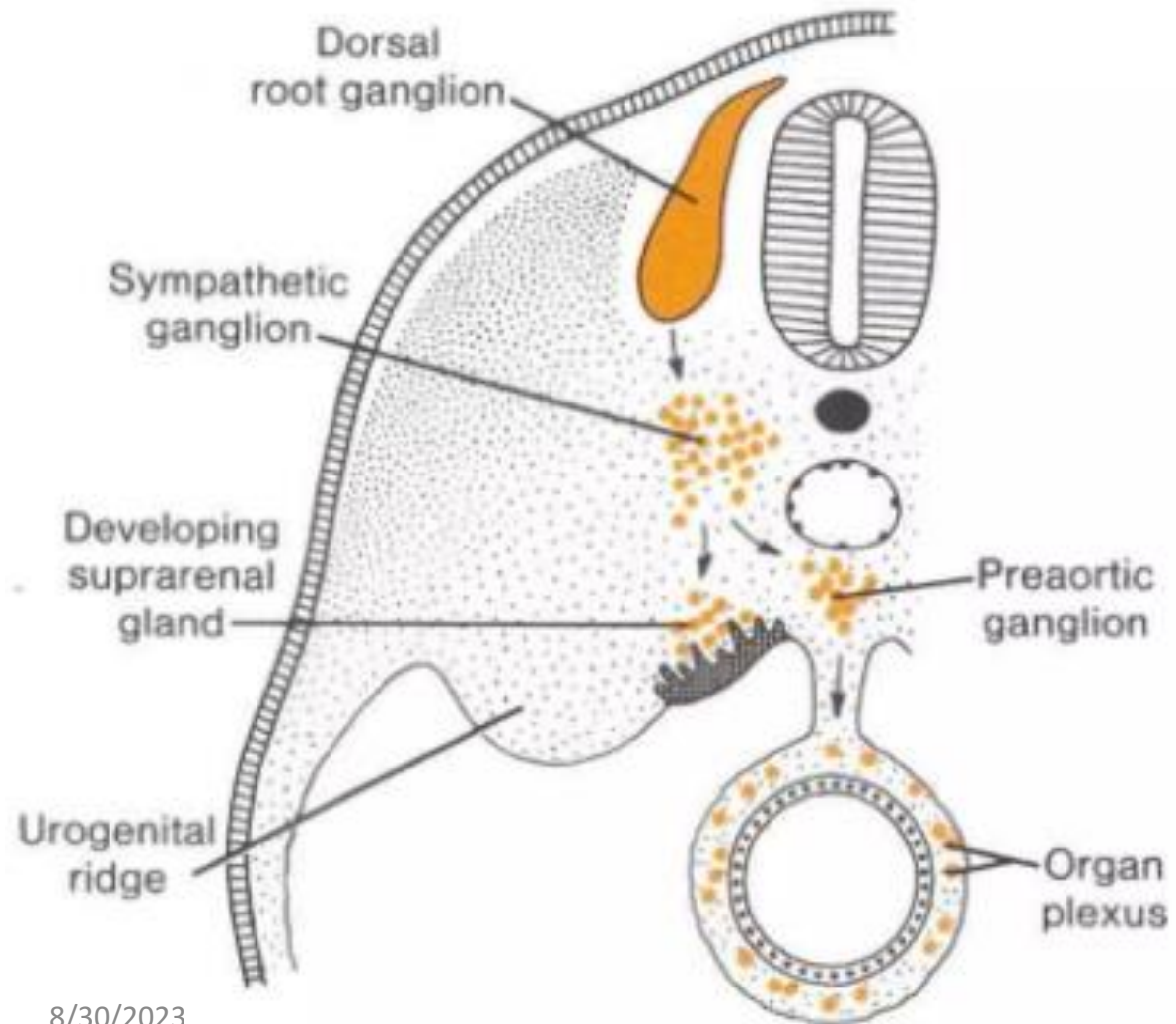
- **Medulla → neural crest cells**

Conn's syndrome is a hormonal condition in which one or both adrenal glands produce more of the hormone aldosterone than normal.

Addison's disease, also called adrenal insufficiency, is an uncommon illness that occurs when the body doesn't make enough of certain hormones.



# Cont....



# Congenital anomalies: Reading assignment

- Pituitary agenesis/hypogenesis
- Ectopic posterior gland
- Pharyngeal hypophysis
- Craniopharygioma
- Lingual thyroid
- Retrosternal thyroid
- Thyroglossal cyst
- Parathyroid agenesis/hypogenesis
- Conn's syndrome & Addison's disease
- Cushing syndrome
- Adrenogenital syndrome
- pheochromocytoma

Cushing's syndrome is a disorder that occurs when your body makes too much of the hormone cortisol over a long period of time.

Adrenogenital syndrome is also known as congenital adrenal hyperplasia, which results from disorders of steroid hormone production in the adrenal glands leading to a deficiency of cortisol.

Pheochromocytoma is a type of neuroendocrine tumor that grows from cells called chromaffin cells

Chromaffin cells are neuroendocrine cells found predominantly in the medulla of the adrenal gland

*Thank  
you*

